A Project Synopsis

On

Smart Document Verification System

For the Degree of
Bachelor of Technology

In

Computer Science and Engineering

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Introduction

In today's digital world, document verification has become very important because digital fraud cases are increasing rapidly. Fake certificates, edited marksheets, manipulated ID cards, and forged signatures are commonly created using advanced image editing tools. Manual verification of documents takes more time and may lead to human errors. Therefore, there is a need for an automated system that can verify documents quickly and accurately.

This project focuses on image preprocessing and multiple feature extraction techniques to detect whether a document is original or tampered. Image preprocessing improves the clarity and quality of the uploaded document image. After preprocessing, important features are extracted using methods such as CNN, SIFT, HOG, LBP, and ORB. These features are then analyzed to classify the document as genuine or fake. The system aims to reduce manual effort and improve verification accuracy.

This project supports the United Nations Sustainable Development Goal (SDG) 9: Industry, Innovation and Infrastructure. The goal promotes innovation and the development of reliable and secure infrastructure. By developing an automated document verification system using advanced image processing and machine learning techniques, this project encourages technological innovation and strengthens digital security infrastructure in institutions such as colleges, banks, and government offices.

Literature Survey

In recent years (2023–2025), document forgery detection systems have become very important because digital fraud is increasing. Many people create fake certificates, edited marksheets, and fake ID cards using advanced image editing software. Because of this, researchers are now focusing on improving document verification systems by using image preprocessing and different feature extraction techniques.

Sharma et al. [1] developed a CNN-based forgery detection system. They used preprocessing steps like converting images to grayscale, improving contrast, and detecting edges before applying CNN. Their research showed that preprocessing improves the accuracy of the model.

Patel and Mehta [2] combined CNN features with HOG features for certificate verification. They found that combining deep learning features with structural features gives better results than using CNN alone.

Kumar et al. [3] used SIFT and ORB techniques to detect important keypoints in documents. They combined these features with deep learning for better authentication. Their system worked well even when documents were rotated or resized.

Li and Wang [4] used LBP (texture-based feature) with CNN for forgery detection. They showed that texture differences help in detecting tampered documents.

Rao et al. [5] used transfer learning with pretrained deep learning models. Their system reduced training time and still gave high accuracy.

Singh and Kaur [6] worked on signature verification using deep CNN and attention mechanisms. Their system detected forged signatures more accurately.

Ahmed et al. [7] used CNN heatmaps to highlight tampered areas in documents. This helps users easily see which part of the document is manipulated.

Chen et al. [8] combined CNN, HOG, and LBP features into one feature vector. They proved that combining multiple features improves reliability and reduces errors.

Gupta and Sharma [9] developed a system that uses preprocessing, feature extraction, and SVM classification. Their system achieved more than 92% accuracy.

Zhang et al. [10] used edge-based analysis with deep learning for government document verification. Their method detected layout and structure changes in documents.

Brown et al. [11] reviewed many recent systems and concluded that hybrid systems (using multiple feature extraction methods) are more accurate and robust than single-method systems.

From recent studies, we understand that image preprocessing is very important because it improves image quality and helps the system work better. It removes noise and makes the image clear so that feature extraction becomes more accurate. CNN is a strong method for extracting deep features from images. But research shows that using only one method is not enough. When traditional feature extraction methods like HOG, SIFT, LBP, and ORB are combined with CNN, the accuracy becomes better. Using multiple features together makes the system more reliable and strong. Many researchers are also working on real-time document verification systems because they are needed in real-life applications. However, most existing systems focus on only one type of document and do not combine all major feature extraction methods in a single system.

This research is important because digital fraud is increasing every day. Modern AI image editing tools have made it easy to create fake documents. Colleges, banks, companies, and government offices need fast and automatic systems to check documents and stop fraud. Studies show that hybrid feature extraction systems give better accuracy than single-method systems. Systems that use multiple features are more secure and dependable. So, building a document verification system that combines image preprocessing with multiple feature extraction techniques is very important today.

Problem Statement

Manual document verification takes a lot of time and can cause human errors. Many fake documents such as edited certificates, marksheets, and ID cards are created using digital tools. Most existing systems cannot properly check the visual details of documents. Therefore, there is a need for an automated system that can verify documents by analyzing image features using preprocessing and multiple feature extraction techniques.

Project Domain

- Image Processing
- Computer Vision
- Deep Learning
- Pattern Recognition
- Document Authentication

Objectives

1. To pre-process uploaded document images using image enhancement techniques in order to improve image clarity and quality.
2. To extract important features from document images using multiple feature extraction methods such as CNN, SIFT, HOG, LBP, and ORB.
3. To compare the extracted features of uploaded documents with genuine document features for verification purposes.
4. To classify the document as original or tampered using an appropriate machine learning or deep learning model.
5. To reduce manual verification time by developing an automated and efficient document verification system.

Proposed Approach / Methodology

Step 1: Image Upload

User uploads document image (JPG/PNG) or PDF.

Step 2: Image Preprocessing

- Resizing
- Noise removal (Gaussian/Median filter)
- Grayscale conversion
- Histogram equalization
- Edge detection (Canny)
- Morphological operations

Purpose: Improve image quality and highlight important structures.

Step 3: Feature Extraction

Multiple feature extraction methods are applied:

1. CNN – Extracts deep features like shapes, layout, signatures, and patterns.
2. SIFT – Detects stable keypoints even if image is rotated or scaled.
3. HOG – Extracts edge direction and structure information.
4. LBP – Captures texture features.
5. ORB – Fast keypoint detection for matching.

All features are combined into a feature vector.

Step 4: Feature Comparison

- Similarity measurement (Cosine similarity / Euclidean distance)
- Classification using Softmax or SVM

Step 5: Output

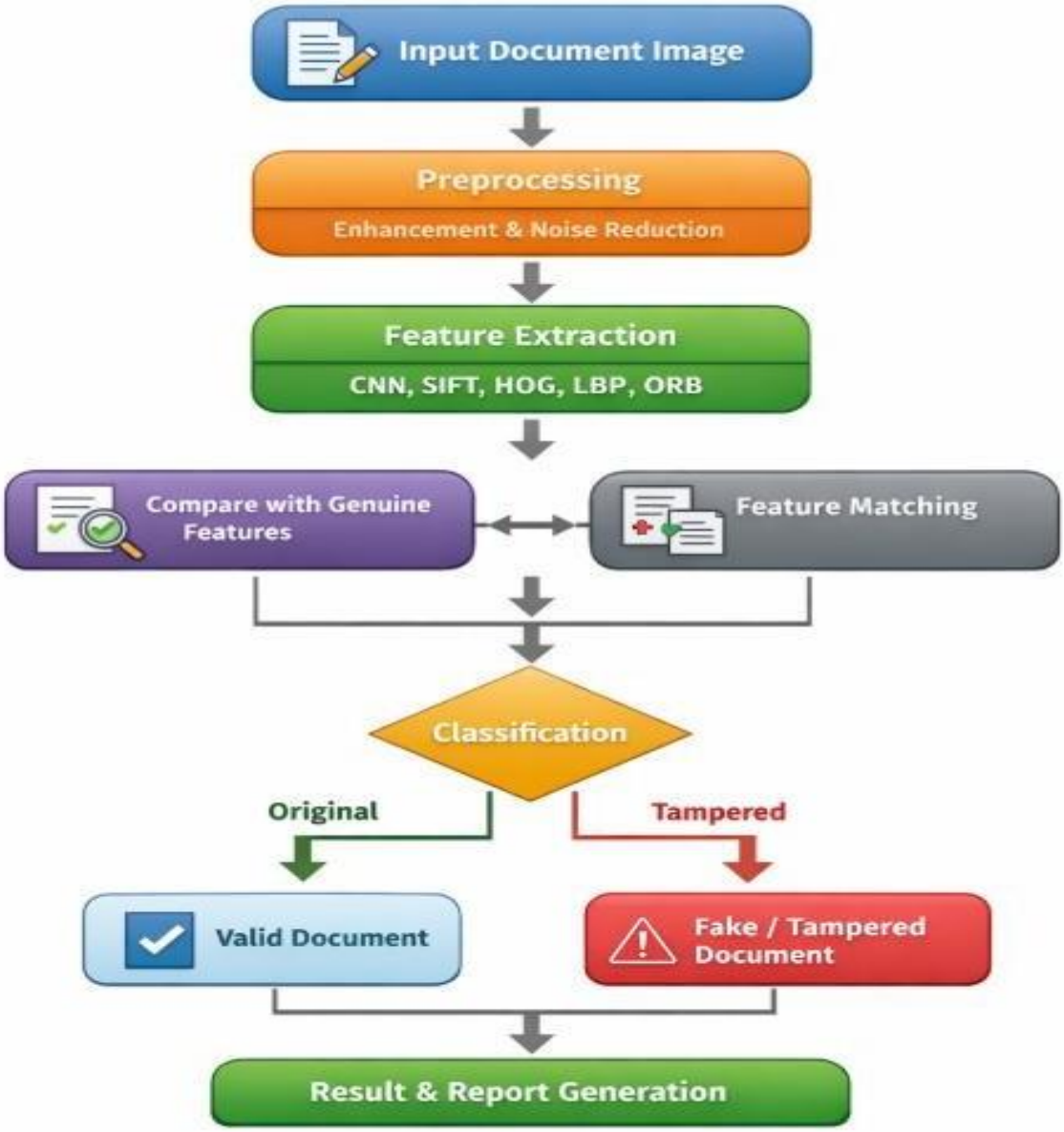
System displays:

- Verified / Not Verified
- Similarity Score
- Confidence Percentage

Proposed / Expected Outcomes / Deliverables

- Automated document verification
- Detection of edited text and signatures
- Reduced manual effort
- Faster processing
- High accuracy system

Flow Chart:



Tech Stack

- Frontend: React
- Backend: Node.js / Python (Flask)
- Image Processing: OpenCV
- Deep Learning: TensorFlow / PyTorch
- Database: MongoDB / MySQL

Project Plan

Future Scopes

- Real-time mobile document verification
- Tampered region detection using heatmaps
- Integration with government portals
- Integration with blockchain for secure storage
- Expansion to passport and banking verification

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